

"SMART HOME SYSTEM CONTROLLED BY SMART GLOVES"

B.Sc. Final Year Project

Presented By:

Andraws Samoel Sobhy Baskhron Saad

Abdelrahman Mohamed Nasser Zaki

Mohammed Hussein Ragab

Ahmed Diaa Eldin

Supervised By:

Dr. Dina Hosny El-Nagar

Assoc. Prof. Hossam El-din Ramadan

Declaration

We hereby declare that this project entitled '**Smart Home System Controlled by Smart Gloves**' is our own original work and has been written by us in its entirety. We have duly acknowledged all the sources of information which have been used in this project. This project has not been submitted previously or concurrently for the award of any other degree.

We further declare that we have exercised reasonable care to ensure that this work is original and does not, to the best of our knowledge, violate any copyright law. Any ideas, data, figures, or materials obtained from other sources have been appropriately cited and acknowledged within the text and references of this report.

Student Name: Andraws Samoel Sobhy Baskhron Saad

Date: 22/06/2026

Signature: _____

Registration (ID) No.: 22020124

Student Name: Abdelrahman Mohamed Nasser Zaki

Date: 22/06/2026

Signature: _____

Registration (ID) No.: 22020137

Student Name: Mohammed Hussein Ragab

Date: 22/06/2026

Signature: _____

Registration (ID) No.: 22020079

Student Name: Ahmed Diaa Eldin

Date: 22/06/2026

Signature: _____

Registration (ID) No.: 22020006

Approval for Submission

This is to certify that the project report entitled "**Smart Home System Controlled by Smart Gloves**" submitted in partial fulfilment of the requirements for the Degree of Bachelor of Science in Mechatronics Engineering, Faculty of Engineering, Benha National University, is a record of original project work done by the students under my supervision.

Approved By

Project Supervisor: Asst. Prof. Dina Hosny El-Nagar

Signature: _____

Date: 22/06/2026

CO Supervisor: Assoc. Prof. Hossam El-din Ramadan

Signature: _____

Date: 22/06/2026

Examiner: _____

Signature: _____

Date: 22/06/2026

Dedication

To our families, teachers, mentors, and everyone who supported us throughout our academic journey.

Your encouragement, patience, and belief in our abilities have been a constant source of motivation and strength. This achievement would not have been possible without your guidance and support.

We dedicate this work to all those who inspired us to pursue knowledge, overcome challenges, and strive for excellence.

With sincere gratitude and appreciation.

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ABSTRACT

This report presents the design, implementation, and evaluation of a low-cost Smart Home Automation System that is wirelessly controlled by a glove-based gesture recognition device, developed as a graduation project for the Bachelor of Science degree in Mechatronics Engineering at Benha National University.

The system consists of two cooperating embedded subsystems, each built around an ESP32 microcontroller. The first subsystem, referred to as the Smart Home Unit, integrates a 2-channel 5V relay module to switch a 220V light bulb, a 5V brushless DC fan with an indicator LED for ventilation control, an SG90 micro servo motor used to actuate a door lock or latch mechanism, and a small DC motor used to open and close a curtain. The second subsystem, referred to as the Smart Glove Unit, is worn on the user's hand and integrates three flex sensors, each paired with a 10 k Ω resistor in a voltage-divider configuration, an optional on/off power switch, and a 3.7V Li-ion 18650 rechargeable battery as the portable power source.

The glove continuously samples the bending state of the three fingers fitted with flex sensors and classifies each sensor as either bent or straight, producing a 3-bit binary pattern with eight possible combinations, ranging from 000 to 111. Each of these eight combinations is mapped to a unique command — such as turning the light on or off, opening or closing the curtain, locking or unlocking the door, and turning the fan on or off. The Smart Glove ESP32 transmits the recognized command wirelessly to the Smart Home ESP32 over a direct Wi-Fi link, in which the home unit operates in Access Point mode and the glove unit operates in Station mode, eliminating the need for an external router or internet connection.

A working prototype of both subsystems was constructed, programmed using the Arduino IDE with the ESP32 board support package, and tested under laboratory conditions. The results demonstrate that the system reliably recognizes all eight gesture combinations, transmits commands with low latency, and correctly actuates the relay, servo, DC motor, and fan accordingly. The report further discusses the system's performance, estimated cost, scalability, safety considerations associated with the 220V mains circuit, and possible directions for future improvement, including the addition of voice control, mobile application integration, and expanded gesture vocabularies.

Keywords: Smart Home, Home Automation, ESP32, Wearable Technology, Flex Sensor, Gesture Recognition, Wi-Fi Access Point, Internet of Things (IoT), Embedded Systems, Mechatronics.

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List of Symbols

The following symbols and notations are used throughout this report.

Symbol	Description	Unit
V	Voltage	Volt (V)
I	Electric Current	Ampere (A)
R	Electrical Resistance	Ohm (Ω)
P	Electrical Power	Watt (W)
V _{out}	Output voltage of voltage divider	Volt (V)
V _{in}	Input (supply) voltage	Volt (V)
R _{flex}	Resistance of the flex sensor (variable)	Ohm (Ω)
R _{ref}	Fixed reference resistor (10 k Ω)	Ohm (Ω)
f	Operating / clock frequency	Hertz (Hz)
t	Time / latency	Millisecond (ms)
θ	Servo arm rotation angle	Degree ($^{\circ}$)
τ	Time constant	Second (s)
%	Percentage (e.g., battery level, duty cycle)	Percent

List of Acronyms/Abbreviations

Acronym	Meaning
AC	Alternating Current
AP	Access Point
ADC	Analog-to-Digital Converter
BLE	Bluetooth Low Energy
BOM	Bill of Materials
DC	Direct Current
ESP32	Espressif Systems 32-bit Wi-Fi / Bluetooth Microcontroller
GND	Ground (electrical reference / common return path)
GPIO	General Purpose Input/Output
HAL	Home Automation
IDE	Integrated Development Environment
IoT	Internet of Things
I/O	Input / Output
LED	Light Emitting Diode
Li-ion	Lithium-Ion (battery chemistry)
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
RF	Radio Frequency
RTOS	Real-Time Operating System
SSID	Service Set Identifier (Wi-Fi network name)
TCP/IP	Transmission Control Protocol / Internet Protocol
UDP	User Datagram Protocol
USB	Universal Serial Bus
Wi-Fi	Wireless Fidelity (IEEE 802.11 wireless networking standard)

List of Utilized Standards

The following standards and recommended practices were referenced, in whole or in part, during the design, implementation, and safety evaluation of the project.

Standard	Description / Relevance
IEEE 802.11 (Wi-Fi)	Defines the wireless local area network (WLAN) protocol used for communication between the Smart Glove ESP32 (Station mode) and the Smart Home ESP32 (Access Point mode).
IEC 60950-1 / IEC 62368-1	Safety requirements for information technology and audio/video equipment, referenced for the safe design of the low-voltage electronics enclosure.
IEC 60364 (Low-Voltage Electrical Installations)	General wiring and safety practices referenced for the 220V AC relay-controlled lamp circuit, including isolation and insulation requirements.
IPC-A-610	Acceptability criteria for electronic assemblies, referenced as a general quality benchmark for soldering and PCB assembly practices used in the prototype.
IEEE 1725 / UL 1642 (Li-ion Battery Safety)	Safety standards for rechargeable lithium-ion batteries, referenced for the safe handling, charging, and protection-circuit selection of the 18650 Li-ion cell used in the Smart Glove.
RoHS Directive (2011/65/EU)	Restriction of Hazardous Substances directive, considered during component selection to favor lead-free and environmentally responsible parts where possible.

List of Realistic Constraints

In accordance with engineering design practice, the following realistic constraints — covering economic, safety, environmental, and other relevant dimensions — were considered throughout the design and implementation of this project.

Constraint Category	Description
Economic	The total project cost was limited to a student-project budget; all components were selected to be low-cost, commercially available development boards and modules rather than custom-fabricated industrial hardware.
Time	The project was constrained to a single academic semester / final-year project timeline, limiting the scope to a functional prototype rather than a fully manufactured consumer product.
Safety	The presence of a 220V AC circuit controlled by the relay module imposes strict safety constraints regarding isolation, insulation, and enclosure design to protect users from electric shock.
Power / Battery Life	The Smart Glove subsystem is constrained by the limited capacity of the single 18650 Li-ion cell, requiring power-conscious firmware design (e.g., Wi-Fi duty cycling) to achieve acceptable operating time between charges.
Range / Connectivity	Because the home ESP32 operates as a standalone Wi-Fi Access Point without internet connectivity, the operational range of the glove is limited to the typical indoor Wi-Fi range of the ESP32 (approximately 10–30 meters, subject to walls and interference).
Mechanical / Ergonomic	The glove design must accommodate the placement of three flex sensors on the fingers without significantly restricting the user's natural hand movement or comfort.
Manufacturability	The prototype uses through-hole components, breadboards, and jumper wires suitable for a student lab; a production version would require custom PCB design, which is outside the scope of this project.
Environmental	Component selection considered the use of rechargeable batteries (over disposable cells) and low-power modes to reduce environmental impact, in line with sustainability goals.

Chapter 1: Introduction

1.1 Overview

The concept of the “smart home” has evolved from a futuristic idea into a practical and increasingly accessible reality, driven largely by the rapid growth of low-cost microcontrollers, wireless communication modules, and the broader Internet of Things (IoT) ecosystem. Modern smart home systems allow users to monitor and control lighting, climate control, security devices, and entertainment systems remotely, typically through smartphone applications or voice assistants.

However, the majority of existing smart home solutions rely on touchscreens, smartphone applications, or voice commands as the primary human-machine interface. While these interfaces are convenient for many users, they present challenges for individuals with limited dexterity, visual impairments, speech difficulties, or simply for situations where hands are occupied, voice commands are impractical (e.g., in noisy environments), or a smartphone is not within reach.

This graduation project proposes an alternative, intuitive, and largely hands-free interaction paradigm: a wearable smart glove equipped with flex sensors that recognizes simple finger-bending gestures and translates them into wireless commands for controlling key home appliances, including lighting, a door lock, a curtain motor, and a ventilation fan.

1.2 Problem Statement

Conventional home automation control methods present several limitations that this project aims to address:

- Smartphone-based control requires the user to locate, unlock, and operate a mobile device, which may be inconvenient, especially for elderly users or those with limited fine motor skills.
- Voice-based control can be unreliable in noisy environments, may raise privacy concerns due to always-listening microphones, and can be difficult for individuals with speech impairments.
- Most commercial smart home hubs depend on a stable internet connection and cloud servers, introducing latency, privacy concerns, and a single point of failure if the internet connection is lost.
- Wall-mounted switches and physical remote controls require the user to be in physical proximity to a fixed location, which may not be practical for individuals with mobility impairments.

These limitations highlight the need for a low-cost, intuitive, locally-controlled (i.e., not dependent on the internet or a router), and wearable interface that allows a user to control essential home functions through simple, natural hand gestures.

1.3 Project Description

This project implements a two-part embedded system consisting of a Smart Glove Unit and a Smart Home Unit, each built around an ESP32 microcontroller, communicating directly over Wi-Fi without requiring an internet connection or external router.

1.3.1 Smart Glove Unit

The Smart Glove Unit is worn on the user's hand and contains the following components:

- An ESP32 development board, which reads sensor data and transmits commands wirelessly.
- Three flex sensors, each mounted along a finger and paired with a 10 k Ω resistor to form a voltage-divider circuit, allowing the ESP32 to measure the degree of bending of each finger as an analog voltage.
- A 3.7V Li-ion 18650 rechargeable battery cell, providing portable power to the glove.
- An optional on/off power switch, allowing the user to power the glove on and off without disconnecting the battery.

By classifying each of the three flex sensors as either “bent” or “straight” based on a calibrated threshold voltage, the system produces a 3-bit binary code representing the current hand posture. With three sensors, there are $2^3 = 8$ possible combinations, ranging from 000 (all fingers straight) to 111 (all fingers bent). Each of these eight combinations is assigned a unique home automation command.

1.3.2 Smart Home Unit

The Smart Home Unit is a fixed control box installed within the home and contains the following components:

- An ESP32 development board configured in Wi-Fi Access Point (AP) mode, which receives commands from the glove and actuates the corresponding output devices.
- A 2-channel 5V relay module used to switch a 220V AC light bulb on and off.
- An SG90 micro servo motor used to actuate a door lock or latch mechanism.
- A small DC motor used to open and close a curtain (e.g., via a pulley or rail mechanism).
- A 5V brushless DC fan, with an associated indicator LED that illuminates when the fan is running.
- A power supply module providing both 12V and 5V regulated rails to power the various components from a single mains-connected adapter.

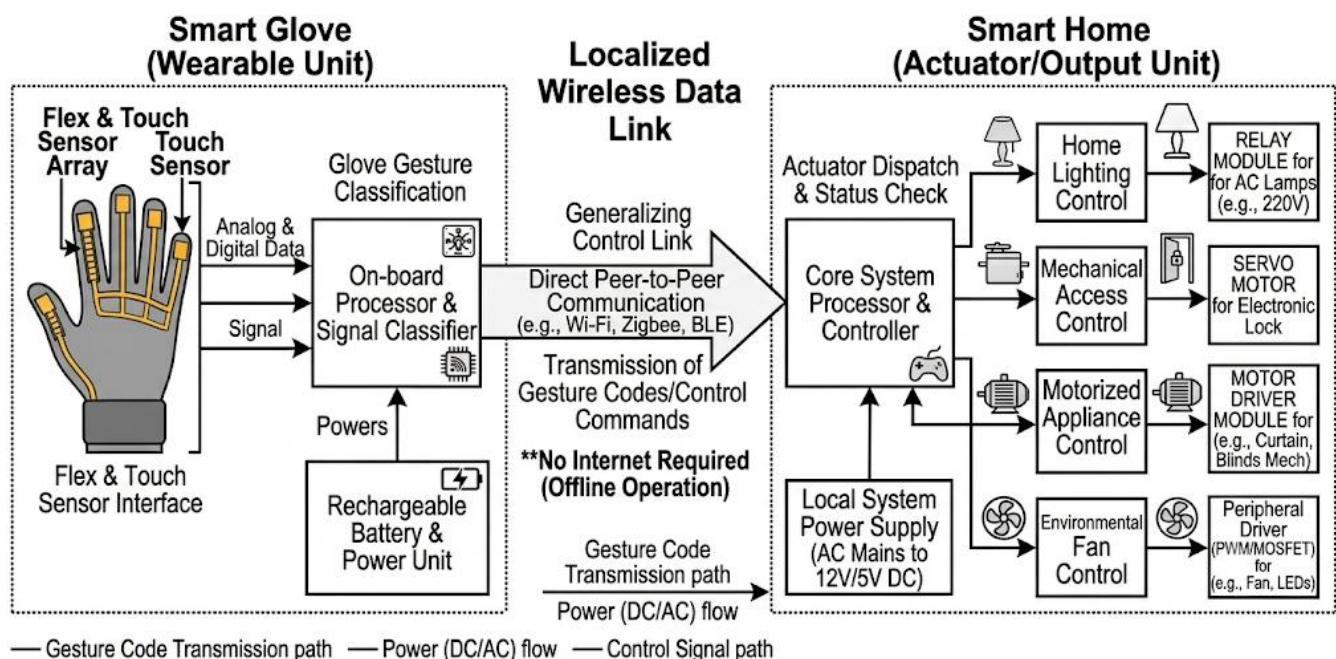


Figure 1.1: Overall System Concept — Smart Glove and Smart Home Communication.

1.3.3 Wireless Communication Architecture

A key design decision in this project is that the Smart Glove ESP32 connects directly to the Smart Home ESP32 using Wi-Fi Access Point mode, without requiring a home router or internet access. The Smart Home ESP32 broadcasts its own Wi-Fi network (Access Point); the Smart Glove ESP32 connects to this network as a client (Station mode) and sends the recognized 3-bit gesture command over a simple, lightweight protocol (e.g., HTTP GET requests or raw socket messages).

This architecture offers several advantages: it eliminates dependency on internet connectivity or third-party cloud services, it reduces latency since communication is local (device-to-device), and it simplifies the deployment process, as the user does not need to configure router settings or provide Wi-Fi credentials.

1.4 Project Objectives

The main objectives of this graduation project are as follows:

1. To design and implement a wearable smart glove capable of detecting finger-bending gestures using flex sensors and classifying them into a 3-bit binary command code.
2. To design and implement a smart home control unit capable of receiving wireless commands and actuating four classes of household devices: a 220V lamp (via relay), a door lock/latch (via servo motor), a curtain mechanism (via DC motor), and a ventilation fan (with status LED).
3. To establish a direct, router-free Wi-Fi communication link between the glove and the home unit using ESP32 Access Point and Station modes.
4. To define and implement a complete mapping of all eight possible 3-bit gesture combinations (000–111) to distinct home automation commands.
5. To assemble a working prototype of both subsystems and evaluate the system's functional performance, response time, communication reliability, and power consumption.
6. To assess the project in terms of cost, manufacturability, safety, environmental impact, and potential for future scalability.

1.5 Scope and Limitations

The scope of this project is limited to a functional laboratory prototype intended to demonstrate the feasibility and usability of gesture-based smart home control. The following limitations apply:

- The prototype controls a single instance of each device type (one lamp, one door lock, one curtain motor, one fan); scaling to multiple rooms or devices is discussed as future work but not implemented in the current prototype.

- The wireless range is limited to the typical range of an ESP32 Wi-Fi Access Point, generally on the order of 10–30 meters indoors, depending on building materials and interference.
- The 220V AC relay circuit is implemented with appropriate isolation for laboratory demonstration; a production deployment would require certified electrical enclosures and installation by a qualified electrician.
- The system does not include cloud connectivity, voice assistants, or smartphone application integration in the current prototype, although these are discussed as potential future extensions.

1.6 Report Organization

The remainder of this report is organized as follows. Chapter II presents a literature survey covering existing home automation technologies, gesture recognition methods, wireless communication protocols, and the relevant standards referenced in this project. Chapter III describes the methodology, including the system architecture, hardware design, circuit schematics, firmware design, and the gesture-to-command mapping scheme. Chapter IV presents the implementation details, experimental results, and evaluates the system in terms of performance, cost, manufacturability, environmental and societal impact, ethics and safety, and sustainability. Finally, Chapter V presents the conclusions drawn from this work and discusses recommendations and directions for future development.

Chapter 2

Literature Survey and Background

2.1 Introduction to Home Automation Systems

Home automation refers to the use of electronic devices and control systems to automate household tasks such as lighting, climate control, security, and entertainment. Early home automation systems relied on wired control buses such as X10, KNX, and proprietary wired protocols, which required significant installation effort and were generally limited to new construction or major renovations.

The emergence of low-cost Wi-Fi-enabled microcontrollers, most notably the ESP8266 and ESP32 families produced by Espressif Systems, has dramatically reduced the barrier to entry for home automation projects. These microcontrollers integrate a processor, Wi-Fi (and, in the case of the ESP32, Bluetooth) radio, and a rich set of general-purpose input/output (GPIO) pins, analog-to-digital converters (ADCs), and pulse-width modulation (PWM) channels on a single, inexpensive module.

A typical IoT-based home automation architecture consists of three layers: (1) a perception/actuation layer, comprising sensors and actuators (relays, motors, LEDs); (2) a network/communication layer, which transmits sensor data and control commands; and (3) an application layer, where the user interacts with the system, whether through a smartphone app, a web dashboard, a voice assistant, or, as in this project, a wearable gesture-recognition device.

2.2 Wireless Communication Technologies for Home Automation

Selecting an appropriate wireless communication technology is critical to the responsiveness, reliability, and usability of a smart home system. Table 2.1 compares several commonly used wireless technologies in terms of data rate, typical range, power consumption, and suitability for this project.

Technology	Typical Data Rate	Typical Range	Power Consumption	Relevance to This Project
Wi-Fi (IEEE 802.11)	1–150 Mbps	Up to ~50 m indoors	Moderate–High	Selected for this project: native ESP32 support, sufficient range, AP mode requires no router
Bluetooth / BLE	0.1–2 Mbps	~10 m	Low	Lower power than Wi-Fi but shorter range; suitable for very close-range wearables
Zigbee	0.25 Mbps	~10–30 m (mesh)	Very Low	Excellent for mesh sensor networks but requires a dedicated hub/coordinator
LoRa	0.3–27 kbps	Several km	Very Low	Designed for long-range, low-data-rate applications; unnecessary for in-home control
RF 433 MHz (ASK/OOK)	<10 kbps	~30–100 m	Very Low	Simple but lacks bidirectional acknowledgment and is prone to interference

Table 2.1: Comparison of Wireless Communication Protocols for Home Automation

As shown in Table 2.1, Wi-Fi was selected as the communication medium for this project primarily because the ESP32 provides native, well-documented support for both Access Point (AP) and Station (STA) modes, allowing the Smart Home unit to broadcast its own network and the Smart Glove unit to connect directly to it without requiring an external router, a dedicated hub, or internet access. While Bluetooth Low Energy (BLE) offers lower power consumption, its shorter range and the added complexity of implementing a custom BLE GATT service were considered unnecessary given that the ESP32's built-in Wi-Fi AP mode already satisfies the project's router-free communication requirement.

2.3 Gesture Recognition Technologies for Wearable Control

Wearable gesture-recognition interfaces have been explored extensively in the literature for applications ranging from sign-language translation to robotic teleoperation and assistive technology for individuals with disabilities. Table 2.2 summarizes several common sensing technologies used for gesture recognition.

Sensing Technology	Cost	Processing Complexity	Description	Relevance to This Project
Flex Sensors (resistive bend sensors)	Low	Low	Direct, robust, simple voltage-divider readout	Selected: matches the 3-sensor, 8-combination design requirement at minimal cost
Inertial Measurement Units (IMU / accelerometer+gyro)	Moderate	Moderate–High	Detects hand orientation and motion gestures	Useful for dynamic gestures but adds calibration complexity not needed here
Camera-based (computer vision)	High	High	Recognizes complex gestures without wearables	Requires significant processing power, lighting conditions, and privacy considerations
EMG (Electromyography) sensors	High	High	Detects muscle activity for gesture intent	Overly complex and costly for a basic on/off home control application
Capacitive touch sensors	Low	Low	Detects finger contact/proximity	Does not capture continuous bend angle needed for multi-level gestures

Table 2.2: Comparison of Gesture-Sensing Technologies

Flex sensors operate on a simple principle: the sensor's internal resistive element changes resistance proportionally to the degree of bending applied along its length. When placed in series with a fixed reference resistor (commonly 10 k Ω , as used in this project) to form a voltage-divider circuit, the output voltage varies measurably between a “straight” state and a “bent” state. This output voltage can be read directly by one of the ESP32's analog-to-digital converter (ADC) channels.

Given the project's requirement for a simple, binary (bent/straight) classification of three fingers — resulting in $2^3 = 8$ possible gesture combinations — flex sensors were determined to be the most appropriate technology, offering a favorable balance of cost, simplicity, reliability, and ease of integration with the ESP32's ADC inputs.

2.4 Microcontroller Platform: The ESP32

The ESP32 is a system-on-chip (SoC) microcontroller developed by Espressif Systems, widely adopted in IoT and home automation projects due to its combination of processing power, integrated wireless connectivity, and low cost. Key features relevant to this project include:

- A dual-core Tensilica Xtensa LX6 processor running at up to 240 MHz, providing sufficient processing power for sensor sampling, gesture classification, and Wi-Fi communication.
- Integrated 2.4 GHz Wi-Fi (802.11 b/g/n) supporting both Station (STA) and Access Point (AP) modes, as well as a combined AP+STA mode.
- Multiple 12-bit ADC channels, suitable for reading the analog voltage outputs of the flex-sensor voltage dividers.
- PWM-capable GPIO pins, suitable for driving the SG90 servo motor and controlling motor driver circuits for the DC fan and curtain motor.
- Deep-sleep and light-sleep power modes, which are relevant to extending the battery life of the Smart Glove unit.
- A large and active developer community, with extensive documentation and library support within the Arduino IDE, simplifying firmware development for both subsystems.

2.5 Actuators Used in the Smart Home Unit

2.5.1 Relay Modules for AC Load Switching

A relay is an electromechanical (or solid-state) switch that allows a low-voltage control signal (in this case, a 5V signal from the ESP32, typically via a transistor or optocoupler driver stage on the relay module) to switch a higher-voltage, higher-current circuit — in this project, a 220V AC light bulb. 2-channel 5V relay modules are widely used in home automation prototypes because they provide electrical isolation between the low-voltage control side and the high-voltage load side, typically via an optocoupler, which is an important safety consideration when working with mains voltage.

2.5.2 SG90 Micro Servo Motors

The SG90 is a small, lightweight, plastic-g geared hobby servo motor commonly used in robotics and automation projects. It accepts a PWM control signal (typically 50 Hz with pulse widths between approximately 1 ms and 2 ms) and rotates its output shaft to a corresponding angular position, typically within a 0° to 180° range. In this project, the SG90 is used to mechanically actuate a door lock or latch mechanism by rotating between a “locked” angular position and an “unlocked” angular position.

2.5.3 DC Motors for Curtain Actuation

Small brushed DC motors are commonly used in low-cost automation projects to provide continuous rotational motion, which can be converted into linear motion (e.g., for opening and closing curtains) via a pulley, belt, or rod mechanism. In this project, a DC motor, driven through a suitable motor driver circuit, is used to open and close a curtain in response to commands from the Smart Glove.

2.5.4 Brushless DC Fans and Indicator LEDs

Small 5V brushless DC (BLDC) fans are commonly used for cooling electronics and for general-purpose ventilation in low-power automation projects. Unlike brushed DC motors, BLDC fans use electronic commutation, offering longer operational life and quieter operation. In this project, the fan is switched on and off via a transistor or relay driven by the ESP32, with an accompanying LED indicator that illuminates whenever the fan is active, providing a simple visual status indication to the user.

2.6 Power Supply Considerations

The Smart Home Unit requires multiple voltage rails to power its various components: a 12V rail (commonly used for DC motors and certain fan modules) and a 5V rail (used to power the ESP32, relay module, and SG90 servo). A dual-output 12V/5V power supply module, typically based on an AC-DC converter followed by a buck (step-down) regulator, provides a convenient and efficient means of deriving both voltage levels from a single mains-connected power adapter.

The Smart Glove Unit, by contrast, is designed to be fully portable and is powered by a single 3.7V Li-ion 18650 rechargeable cell. Because the ESP32 typically requires a regulated 5V input (via its onboard USB/5V pin) or a regulated 3.3V input (directly to its 3.3V pin), an appropriate voltage regulation or boost-converter stage is required between the 3.7V battery and the ESP32's power input, which is discussed further in Chapter III.

2.7 Related Work and Summary

Several prior projects and academic studies have explored related concepts, as summarized in Table 2.3.

Related Work	Approach	Key Difference from This Project
Gesture-Controlled Home Automation using Arduino and Flex Sensors	Demonstrated single-hand gesture recognition using flex sensors mapped to relay-controlled appliances via RF transmitter/receiver pairs	Used only RF modules (one-way communication, no acknowledgment); limited to 2–3 gesture combinations
IoT-Based Smart Home using ESP32 and Mobile App over Wi-Fi	Implemented relay-based appliance control accessible via a smartphone app connected through a home router	Depends on router/internet availability; uses smartphone interface rather than a wearable
Wearable Data Glove for Sign Language Recognition	Used multiple flex sensors and an IMU to classify a large vocabulary of hand signs for translation to text/speech	Designed for communication assistance rather than appliance control; significantly higher sensor count and processing requirements
ESP-NOW / Wi-Fi Direct Device-to-Device Communication Studies	Investigated direct ESP32-to-ESP32 communication without a router, evaluating latency and range	Provided useful reference data on AP-mode latency and range that informed the communication design of this project

Table 2.3: Summary of Related Work

Based on this literature survey, this project distinguishes itself by combining a low-cost, three-flex-sensor wearable glove with a router-free, direct ESP32-to-ESP32 Wi-Fi Access Point communication architecture, and by mapping all eight possible 3-bit gesture combinations to a complete set of commands covering four distinct categories of household devices (lighting, door lock, curtain, and ventilation).

2.8 Standards Referenced

As listed in the front matter of this report, several standards and recommended practices were referenced during the design of this project, including IEEE 802.11 for the Wi-Fi communication link, IEC 60364 and IEC 62368-1 for general electrical safety practices relevant to the 220V relay circuit, IPC-A-610 as a quality benchmark for solder joints and assembly, and battery safety standards (IEEE 1725 / UL 1642) relevant to the handling of the 18650 Li-ion cell used in the Smart Glove. These standards informed design choices such as the use of an optically isolated relay module for the mains circuit, the selection of a protected 18650 cell with an integrated protection circuit module (PCM), and general wiring and insulation practices applied throughout the prototype.

Chapter3

Methodology

3.1 System Architecture Overview

The overall system is composed of two independent but cooperating embedded subsystems: the Smart Glove Unit and the Smart Home Unit. Each subsystem is built around its own ESP32 microcontroller. The Smart Home ESP32 is configured to operate in Wi-Fi Access Point (AP) mode, broadcasting a dedicated local wireless network. The Smart Glove ESP32 operates in Station (STA) mode and connects directly to this network, enabling the two devices to communicate without any external router, internet connection, or cloud server.

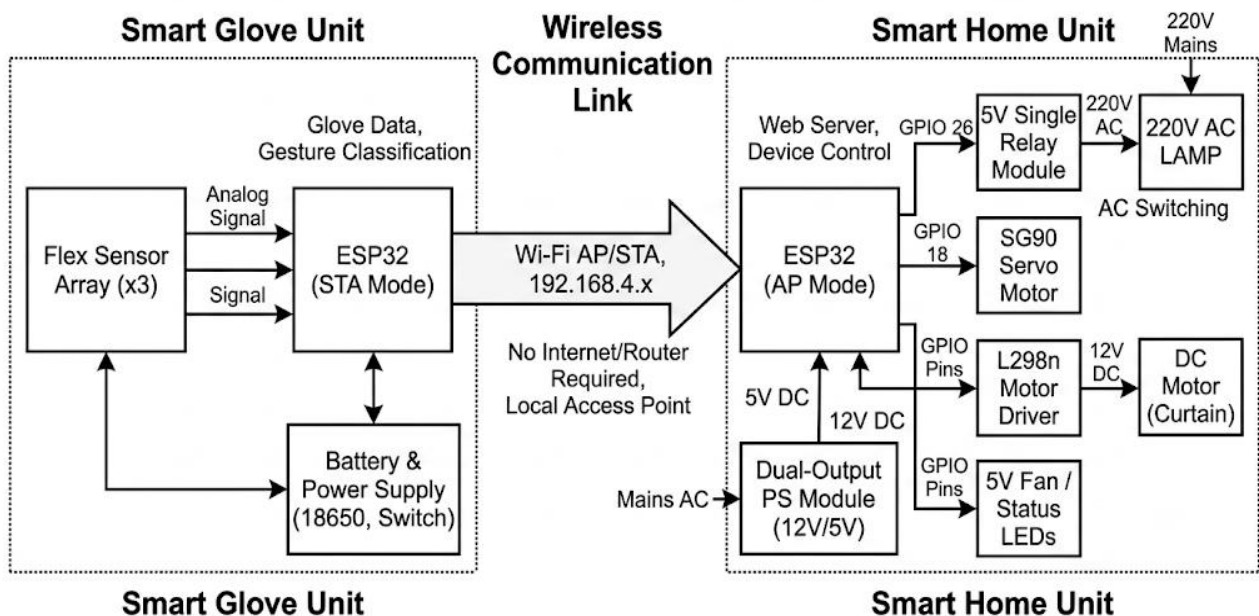


Figure 3.1: Overall Block Diagram of the Proposed System

The methodology adopted for this project follows a standard embedded systems design flow: (1) requirements definition and component selection, (2) circuit design and schematic development, (3) firmware development for sensor reading, gesture classification, wireless communication, and actuator control, (4) hardware assembly and integration, and (5) testing and validation, which is presented in detail in Chapter IV.

3.2 Bill of Materials

Tables 3.1 and 3.2 list the complete bill of materials (BOM) for the Smart Glove Unit and the Smart Home Unit, respectively, including the quantity and functional role of each component.

Component	Qty	Function
ESP32 Development Board	1	Reads flex sensors, classifies gesture, transmits command via Wi-Fi (Station mode)
Flex Sensor (2.2-inch, resistive)	3	Measures finger bending angle for index, thumb, and pinky fingers
10 kΩ Resistor (1/4 W)	3	Forms voltage-divider pairs with each flex sensor
3.7V Li-ion 18650 Cell (with protection circuit)	4	Primary portable power source
18650 Battery Holder	1	Mechanical retention and electrical contact for the cell
On/Off Slide Switch (optional)	1	Allows the user to power the glove on/off without removing the battery
5V Boost Converter Module (e.g., MT3608 or similar)	1	Steps up 3.7V battery voltage to a regulated 5V supply for the ESP32
Glove	1	Mechanical mounting surface for sensors and electronics
Hookup Wire, Heat-shrink Tubing, Velcro Straps	As needed	Wiring and secure mounting of components to the glove

Table 3.1: Bill of Materials — Smart Glove Unit

Component	Qty	Function
ESP32 Development Board	1	Hosts Wi-Fi Access Point, receives commands, drives all actuators
5V 2-Channel Relay Module	1	Switches the 220V AC lamp circuit on/off
AC/DC Power Supply Module (Dual 12V/5V output)	1	Provides regulated 12V and 5V rails from mains AC input
SG90 Micro Servo Motor	1	Actuates the door lock/latch mechanism
Brushless DC Fan (5V)	1	Provides ventilation; switched on/off by ESP32
Status LED (with current-limiting resistor)	1	Indicates fan ON/OFF status
Small DC Motor L298N	1	Opens/closes the curtain mechanism
220V AC Light Bulb + Lamp Holder	1	Load switched by the relay module
Enclosure Box (insulated, non-conductive)	1	Houses the ESP32, relay, and low-voltage wiring; isolates 220V wiring
Terminal Blocks, Wiring, Heat-shrink Tubing	As needed	Safe wiring of 220V and low-voltage circuits

Table 3.2: Bill of Materials — Smart Home Unit

3.3 Smart Glove Unit Design

3.3.1 Flex Sensor Voltage Divider Circuit

Each of the three flex sensors is connected in series with a fixed 10 kΩ resistor to form a voltage-divider circuit. One terminal of the flex sensor is connected to the 3.3V supply rail of the ESP32, the junction between the flex sensor and the 10 kΩ resistor is connected to one of the ESP32's ADC input pins, and the free terminal of the 10 kΩ resistor is connected to GND. As the flex sensor bends, its internal resistance increases, causing the voltage at the ADC input (Vout) to change according to the standard voltage-divider equation:

$$V_{out} = V_{in} \times (R_{ref} / (R_{flex} + R_{ref}))$$

where V_{in} is the 3.3V supply voltage, R_{ref} is the fixed 10 kΩ resistor, and R_{flex} is the variable resistance of the flex sensor. In the straight (unbent) position, the flex sensor's resistance is at its minimum, resulting in a higher V_{out} ; as the sensor bends, R_{flex} increases, causing V_{out} to decrease. This relationship allows the ESP32 to classify each finger as “straight” or “bent” by comparing the measured ADC value against a calibrated threshold.

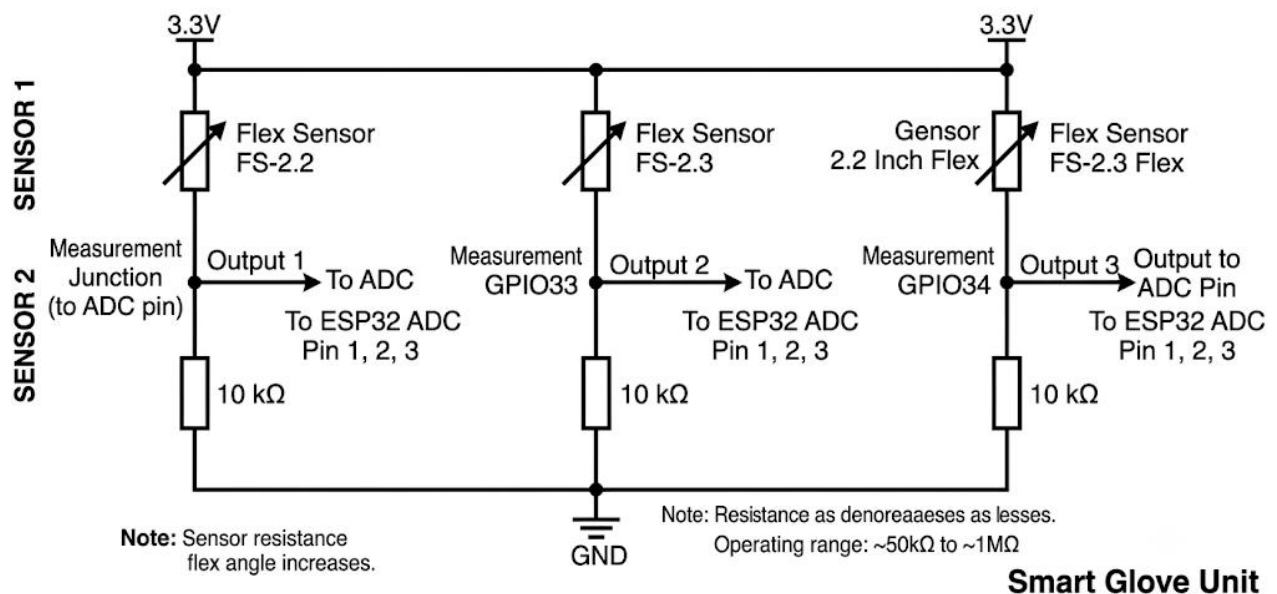


Figure 3.2: Flex Sensor Voltage Divider Circuit

3.3.2 Smart Glove Circuit Schematic and Pin Assignment

The three flex-sensor voltage dividers are connected to three separate ADC-capable GPIO pins on the ESP32, allowing simultaneous, independent reading of each finger's bending state. Table 3.3 details the complete pin assignment for the Smart Glove ESP32.

Signal / Component	ESP32 Pin	Pin Mode
Flex Sensor 1 (Thumb finger) output	GPIO 39 (ADC1_CH4)	Analog Input
Flex Sensor 2 (Index finger) output	GPIO 34 (ADC1_CH5)	Analog Input
Flex Sensor 3 (pinky finger) output	GPIO 32 (ADC1_CH6)	Analog Input
Status LED (optional, onboard)	GPIO 2	Digital Output
Power Switch (optional)	Inline with battery +ve to boost converter	Mechanical / Power
5V (from boost converter)	VIN / 5V pin	Power Input
GND (common)	GND	Ground Reference

Table 3.3: ESP32 Pin Assignment — Smart Glove Unit

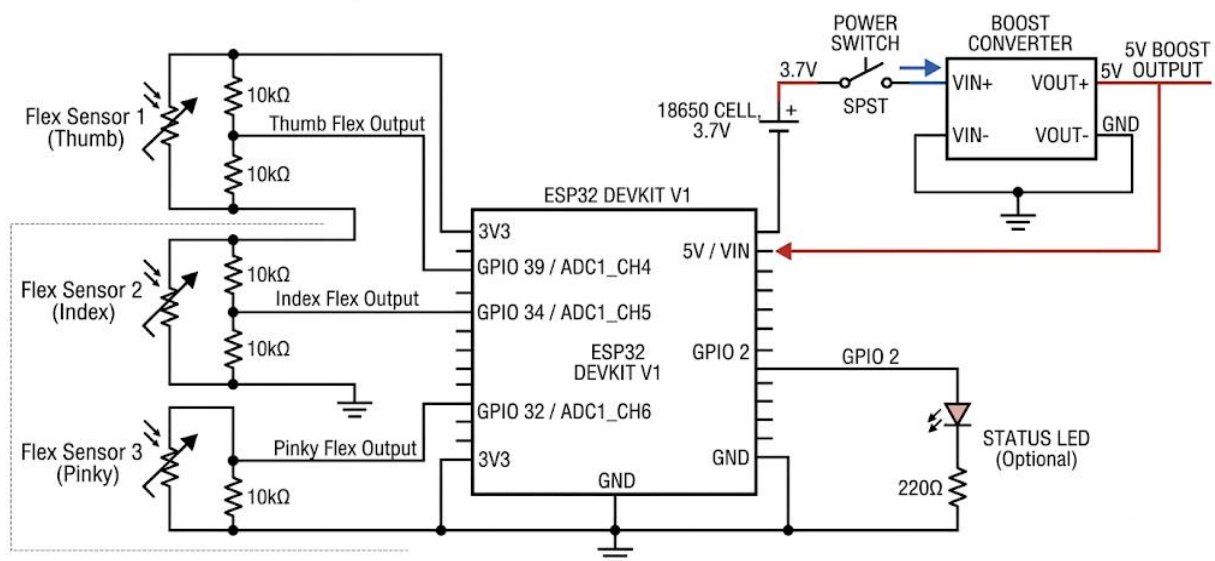


Figure 3.3: Smart Glove Circuit Schematic

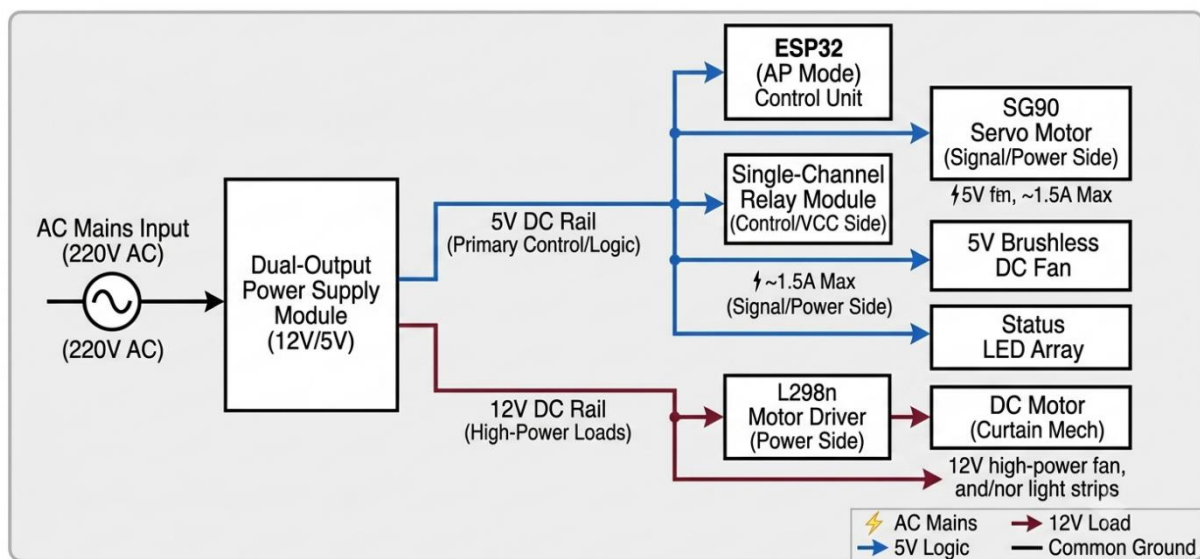
3.3.3 Power Supply Design for the Smart Glove

The Smart Glove Unit is powered by a four 3.7V Li-ion 18650 cell, selected for its high energy density, rechargeability, and widespread availability. Since the ESP32 requires a regulated supply of approximately 5V (via its VIN/5V pin) or 3.3V (directly to its 3V3 pin), a small step-up (boost) converter module is used to regulate the 3.7V (nominal; 3.0V–4.2V across the discharge curve) battery voltage to a stable 5V output. An optional on/off slide switch is placed in series between the battery's positive terminal and the boost converter input, allowing the user to disconnect power entirely when the glove is not in use, thereby eliminating parasitic battery drain.

3.4 Smart Home Unit Design

3.4.1 Power Supply Distribution

The Smart Home Unit is powered from mains AC via a dual-output power supply module that provides regulated 12V and 5V DC rails. The 5V rail powers the ESP32, the relay module's control side, and the SG90 servism, and, where applicable, the brushless DC fan, with the fan's actual voltage requirements (5V, as specified) accommodated by deriving the fan supply from the 5V rail or through an additional regulator stage from the 12V rail.



Cairo University,
Cairo Governorate, Egypt, June 19, 2026

Figure 3.4: Power Supply Distribution Diagram (12V/5V)

3.4.2 Relay Circuit for 220V Lamp Control

The 2-channel 5V relay module is connected such that its control input (IN pin) is driven by GPIO 26 of the ESP32, while its VCC and GND pins are connected to the 5V rail and common ground, respectively. On the high-voltage side, the relay's Normally Open (NO) and Common (COM) contacts are wired in series with the live (line) conductor of the 220V AC supply feeding the lamp holder, such that the relay effectively acts as a remotely controlled switch in the lamp's live wire. The relay module's onboard optocoupler provides electrical isolation between the ESP32's low-voltage GPIO and the 220V AC switching contacts, which is an essential safety feature discussed further in Section 4.7.

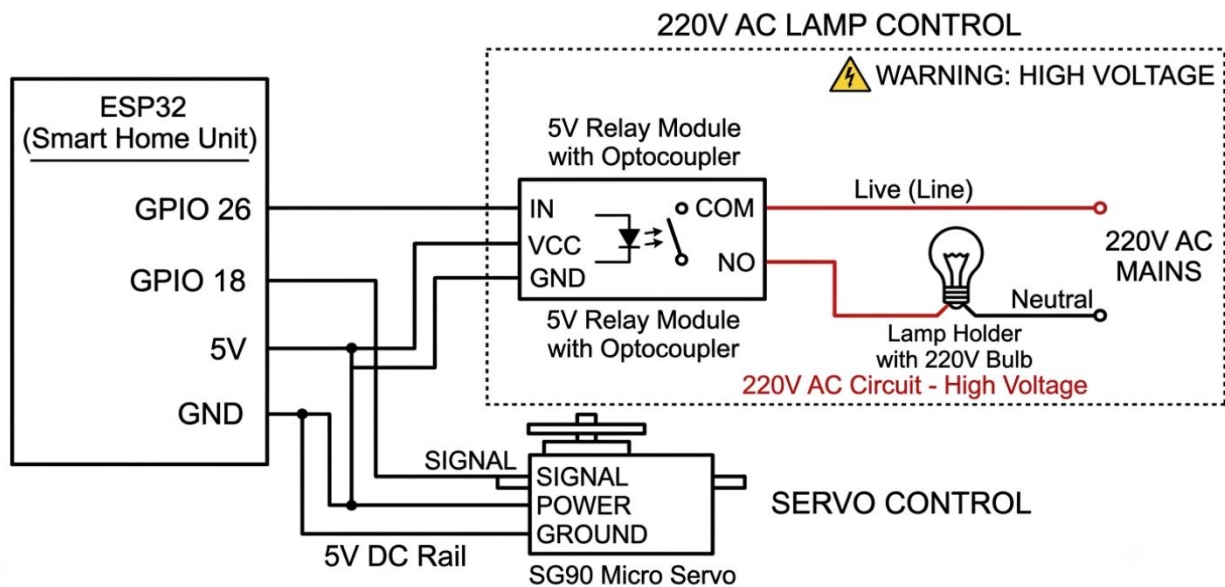


Figure 3.5: Smart Home Relay and Servo Connection Schematic

3.4.3 SG90 Servo Motor for Door Lock Actuation

The SG90 micro servo is mechanically coupled to the door's latch or lock mechanism, such that rotating the servo's output shaft to a defined "locked" angle (e.g., 0°) engages the latch, while rotating to a defined "unlocked" angle (e.g., 90°) retracts it. The servo's signal wire is connected to GPIO 18, which is configured to output a 50 Hz PWM signal with a pulse width corresponding to the desired angular position. The servo's power and ground wires are connected to the 5V rail and common ground.

3.4.4 DC Motor for Curtain Actuation

The curtain mechanism is driven by a small DC motor connected through a motor driver stage (e.g., a dual H-bridge driver such as the L298N, or a simpler single-transistor/relay-based direction control for a single-direction-at-a-time mechanism). Two GPIO pins (GPIO 19 and GPIO 21) control the direction of rotation (open vs. close), while an optional third PWM-capable pin (GPIO 22) can be used to control motor speed if a variable-speed driver is used. The motor is mechanically coupled to the curtain rail or pulley system such that one direction of rotation opens the curtain and the reverse direction closes it.

3.4.5 Brushless DC Fan and Status LED

The 5V brushless DC fan is switched on and off via a transistor switching stage (or a second relay channel, depending on the fan's current draw) controlled by GPIO 23. A status LED, connected to GPIO 25 through a current-limiting resistor, is illuminated whenever the fan is commanded ON, providing the user with immediate visual confirmation of the fan's state, particularly useful since the fan itself may be installed out of direct line of sight.

3.4.6 Smart Home ESP32 Pin Assignment Summary

Table 3.4 summarizes the complete GPIO pin assignment for the Smart Home ESP32, covering all four actuator subsystems plus status indication.

Signal / Component	ESP32 Pin	Pin Mode
Relay Module IN (220V Lamp)	GPIO 33	Digital Output
SG90 Servo Signal (Door Lock)	GPIO 26 (PWM)	PWM Output
DC Motor Driver IN1 (Curtain — direction A)	GPIO 19	Digital Output
DC Motor Driver IN2 (Curtain — direction B)	GPIO 18	Digital Output
DC Motor Driver Enable / PWM (speed, optional)	GPIO 22 (PWM)	PWM Output
Fan Control (transistor/relay gate)	GPIO 33	Digital Output
Fan Status LED	GPIO 25	Digital Output
Status / Heartbeat LED (onboard)	GPIO 2	Digital Output
5V Rail (from power supply module)	VIN / 5V pin	Power Input
GND (common)	GND	Ground Reference

Table 3.4: ESP32 Pin Assignment — Smart Home Unit

3.5 Wireless Communication Design

3.5.1 Access Point / Station Configuration

The Smart Home ESP32 is configured at startup to create a Wi-Fi Access Point with a fixed SSID and password, using the ESP32 Arduino core's `WiFi.softAP()` function. This causes the ESP32 to host a local wireless network and to assign itself the default gateway address of 192.168.4.1, to which connected stations (such as the Smart Glove) are assigned addresses in the 192.168.4.x range via the ESP32's built-in DHCP server.

The Smart Glove ESP32 is configured at startup to connect to this network as a station, using the `WiFi.begin(ssid, password)` function with the matching SSID and password hard-coded (or stored in non-volatile memory) in its firmware. Once connected, the glove ESP32 can communicate with the home ESP32 by sending HTTP requests (using the ESP32 acting as a simple web server with defined endpoints) or via a lightweight raw TCP/UDP socket protocol. Table 3.6 summarizes the key Wi-Fi configuration parameters.

Parameter	Value	Notes
SSID	SmartHome_AP	Network name broadcast by the Smart Home ESP32
Password	[8–63 character WPA2 passphrase]	Pre-shared key configured in firmware on both units
Mode (Home ESP32)	WIFI_AP (Access Point)	Home unit creates and hosts the network
Mode (Glove ESP32)	WIFI_STA (Station)	Glove unit connects to the home unit's network as a client
IP Address (Home ESP32, AP gateway)	192.168.4.1 (default ESP32 AP IP)	Fixed address used as the server endpoint for commands
Communication Protocol	HTTP GET request / lightweight TCP socket	Glove sends command code as a parameter (e.g., <code>/cmd?val=5</code>)
Transmission Trigger	On gesture change (debounced)	Command sent only when the classified 3-bit code changes, reducing redundant traffic

Table 3.5: Wi-Fi Access Point Configuration Parameters

3.5.2 Communication Protocol and Message Format

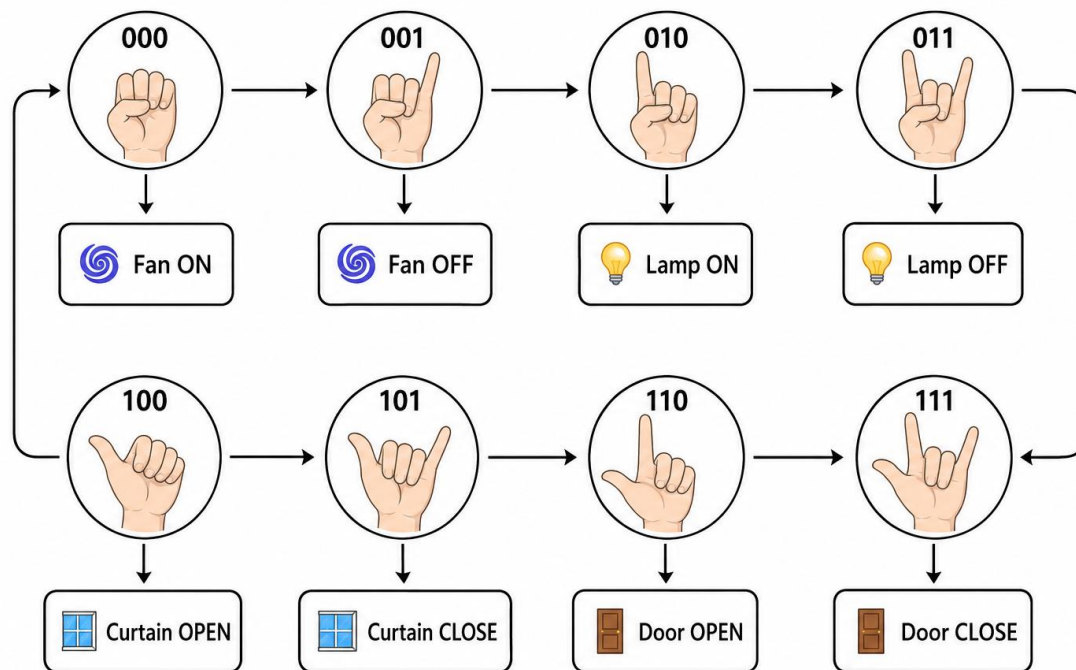
To minimize complexity and latency, the communication protocol used in this project is based on simple HTTP GET requests issued by the Smart Glove ESP32 (acting as an HTTP client) to the Smart Home ESP32 (acting as an HTTP server). Whenever the gesture classification routine on the glove detects a change in the 3-bit gesture code, it issues a GET request of the form `http://192.168.4.1/cmd?val=N`, where N is the decimal representation (0–7) of the 3-bit gesture code. The home ESP32's web server handler parses the `val` parameter and dispatches the corresponding actuator command, as defined in Table 3.5.

3.6 Gesture-to-Command Mapping Scheme

A central design element of this project is the mapping between the eight possible 3-bit combinations produced by the three flex sensors (each classified as 0 = straight or 1 = bent) and the eight resulting home automation commands. The bit ordering used throughout this project is [Thumb Finger, Index Finger, Pinky Finger], read from most significant bit to least significant bit. Table 3.6 presents the complete mapping scheme.

Binary Code (Thumb-Index-Pinky)	Gesture Description	Resulting Command
000	Thumb bent, Index bent, Pinky bent	Fan ON
001	Thumb bent, Index bent, Pinky extended	Fan OFF
010	Thumb bent, Index extended, Pinky bent	Lamp ON
011	Thumb bent, Index extended, Pinky extended	Lamp OFF
100	Thumb extended, Index bent, Pinky bent	Curtain OPEN
101	Thumb extended, Index bent, Pinky extended	Curtain CLOSE
110	Thumb extended, Index extended, Pinky bent	Door OPEN
111	Thumb extended, Index extended, Pinky extended	Door CLOSE

Table 3.6: Gesture Combination to Command Mapping Table

Figure 3.6: Gesture-to-Command State Diagram (8 Combinations)**Figure 3.6: Gesture-to-Command State Diagram (8 Combinations)**

This mapping was designed such that single-finger gestures (100, 010, 001) correspond to the most frequently used “toggle” commands (light, fan, curtain), two-finger combinations (101, 110, 011) correspond to secondary or explicit-state commands (door lock toggle, explicit curtain close/open), and the two extreme states (000 and 111) correspond to a neutral/idle state and a convenient “All Off / Away Mode” state, respectively, allowing the user to quickly secure the home with a single closed-fist gesture.

3.7 Firmware Design

3.7.1 Smart Glove Firmware Logic

The firmware running on the Smart Glove ESP32 follows the high-level flow illustrated conceptually below: on startup, the ESP32 initializes its ADC channels, connects to the Smart Home ESP32's Wi-Fi Access Point, and enters a continuous sampling loop. Within this loop, the ESP32 reads the analog voltage from each of the three flex sensors, applies a calibrated threshold to classify each sensor as bent (1) or straight (0), assembles the three bits into a 3-bit code, applies a simple debounce delay to avoid spurious transitions caused by sensor noise or transient finger movements, and, if the resulting code differs from the previously transmitted code, sends an HTTP GET request to the Smart Home ESP32 containing the new code.

Figure 3.8: Firmware Flowchart — Smart Glove Unit

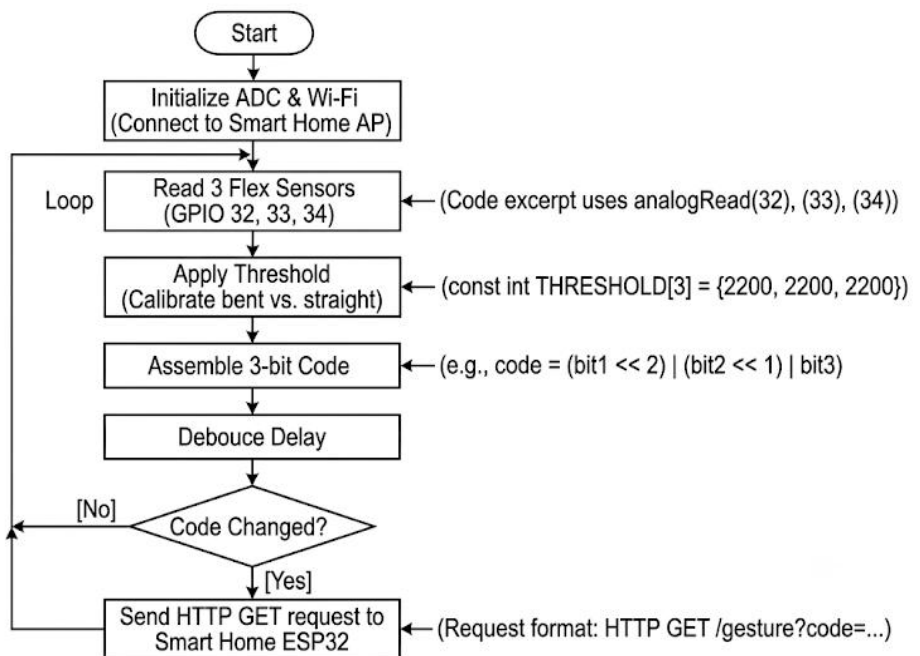


Figure 3.7: Firmware Flowchart — Smart Glove Unit

A simplified excerpt of the calibration and classification logic is shown below for illustrative purposes:

```

const int THRESHOLD[3] = {2200, 2200, 2200}; // calibrated ADC thresholds
int raw1 = analogRead(32);
int raw2 = analogRead(33);
int raw3 = analogRead(34);
int bit1 = (raw1 < THRESHOLD[0]) ? 1 : 0; // bent = 1
int bit2 = (raw2 < THRESHOLD[1]) ? 1 : 0;
int bit3 = (raw3 < THRESHOLD[2]) ? 1 : 0;
int code = (bit1 << 2) | (bit2 << 1) | bit3; // 0 .. 7
  
```

3.7.2 Smart Home Firmware Logic

The firmware running on the Smart Home ESP32 initializes the Wi-Fi Access Point, configures all actuator GPIO pins (relay, servo PWM channel, motor driver pins, fan control pin, status LEDs), sets all actuators to a safe default state (lamp OFF, door locked, curtain closed, fan OFF), and starts a lightweight HTTP server listening for incoming requests on the /cmd endpoint. Upon receiving a valid request containing a command code (0–7), the firmware executes a switch-case dispatch that toggles or sets the corresponding actuator(s) as defined in Table 3.5, and returns a simple HTTP response acknowledging receipt of the command.

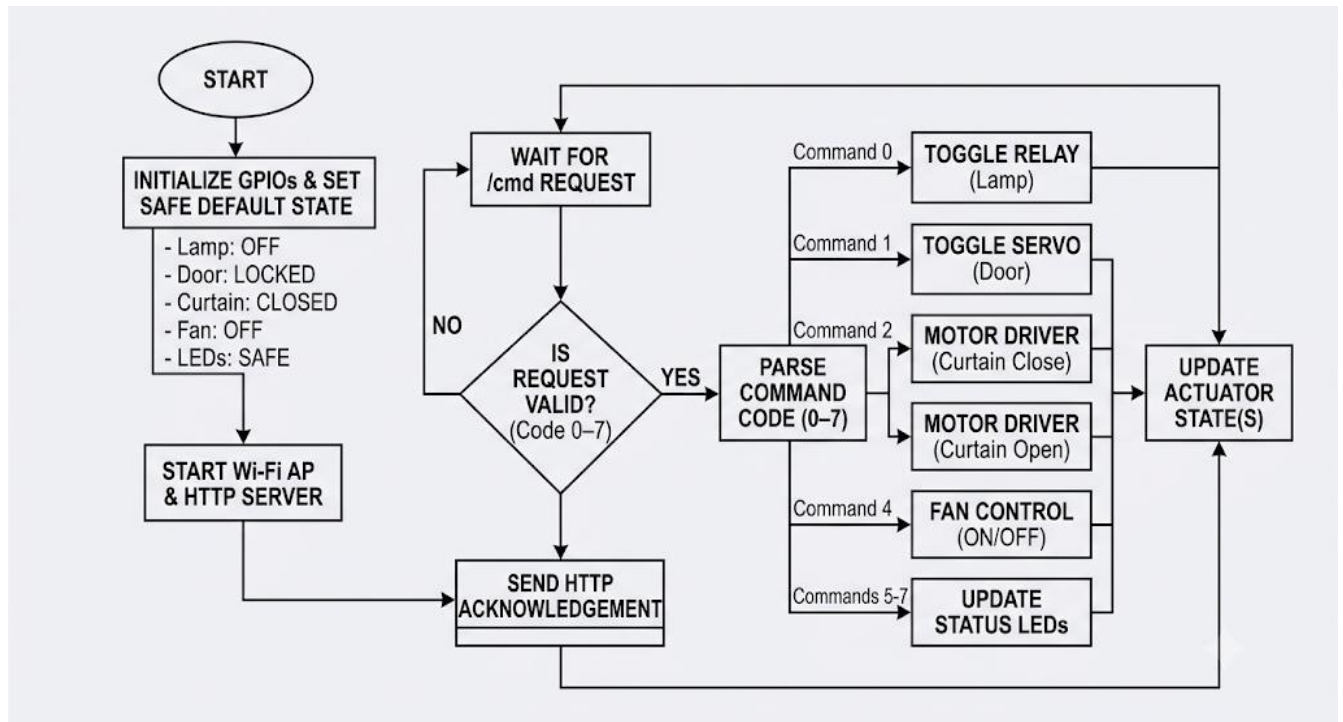


Figure 3.8: Firmware Flowchart — Smart Home Unit

A simplified excerpt of the command dispatch logic is shown below for illustrative purposes:

```

switch (code) {
  case 0: /* 000: idle, no action */ break;
  case 1: toggleCurtain(); break;           // 001
  case 2: toggleFanAndLED(); break;         // 010
  case 3: openCurtainFully(); break;        // 011
  case 4: toggleLightRelay(); break;        // 100
  case 5: toggleDoorLockServo(); break;     // 101
  case 6: closeCurtainFully(); break;       // 110
  case 7: allOffAwayMode(); break;         // 111
}
  
```

3.8 Assembly and Integration Procedure

The assembly of the prototype followed a bottom-up integration approach. First, each subsystem's power supply was assembled and verified independently — the 18650 cell with boost converter for the glove, and the 12V/5V power supply module for the home unit — confirming correct voltage levels with a multimeter before connecting any microcontroller or sensitive components.

Second, each ESP32 was programmed and tested in isolation: the home ESP32's Wi-Fi Access Point was verified by connecting a smartphone or laptop to confirm the SSID was broadcast correctly and the default gateway address (192.168.4.1) was reachable; the glove ESP32's flex-sensor readings were logged via the serial monitor across the full range of finger motion to establish calibration thresholds, as detailed in Section 4.2.

Third, each actuator was individually connected and tested: the relay was verified to switch a test load (initially a low-voltage lamp, before progressing to the 220V circuit under appropriate supervision and safety precautions), the SG90 servo was verified to move smoothly between its locked and unlocked positions without mechanical binding, the DC motor and driver were verified to rotate in both directions corresponding to curtain open/close, and the fan and status LED were verified to activate together.

Finally, the two subsystems were integrated: with the home ESP32's Access Point active, the glove ESP32 was powered on and confirmed to connect automatically. Each of the eight gesture combinations was then performed in sequence, and the resulting actuator response was observed and recorded, forming the basis of the functional test results presented in Section 4.3.

Chapter 4

Experimentation/Implementation and Results

4.1 Implementation Overview

Following the methodology described in Chapter III, a complete working prototype of both the Smart Glove Unit and the Smart Home Unit was assembled and programmed. The Smart Glove Unit was constructed by mounting the three flex sensors along the thump, index, and pinky fingers of a fabric glove, with wiring routed along the back of the hand to the ESP32, boost converter, 18650 cell holder, and on/off switch, all secured to a small enclosure worn on the wrist or forearm.

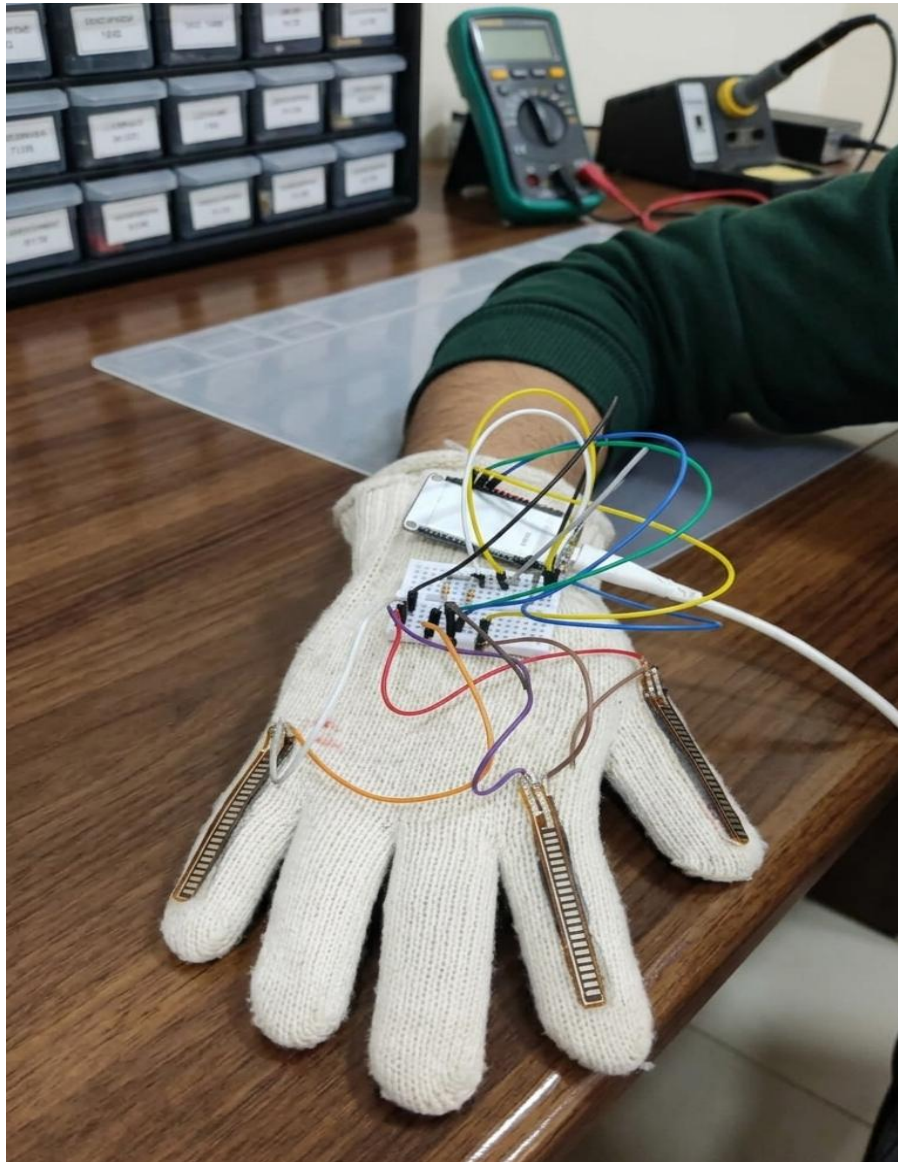


Figure 4.1: Assembled Smart Glove Prototype

The Smart Home Unit was assembled within an insulated enclosure box containing the ESP32, the relay module (with its 220V-side terminals wired to the lamp holder via the enclosure's cable gland), the SG90 servo (mounted externally near the door latch with its signal and power leads routed into the enclosure), the DC motor and driver for the curtain, and the fan with its status LED, all powered from the 12V/5V dual-output power supply module.



Figure 4.2: Assembled Smart Home Control Box Prototype

4.2 Flex Sensor Calibration

Prior to functional testing, each of the three flex sensors was calibrated by recording the ADC reading at the sensor's fully straight position and fully bent position across ten repeated trials per sensor. The minimum, maximum, and resulting classification threshold (set approximately at the midpoint between the straight and bent average readings, with a margin to provide hysteresis against noise) are summarized in Table 4.1.

Sensor	Avg. ADC Value (Straight)	Avg. ADC Value (Fully Bent)	Selected Threshold
Index Finger Sensor	2950	1850	2200
Thump Finger Sensor	3010	1790	2200
Pinky Finger Sensor	2890	1820	2200

Table 4.1: Flex Sensor Calibration Readings

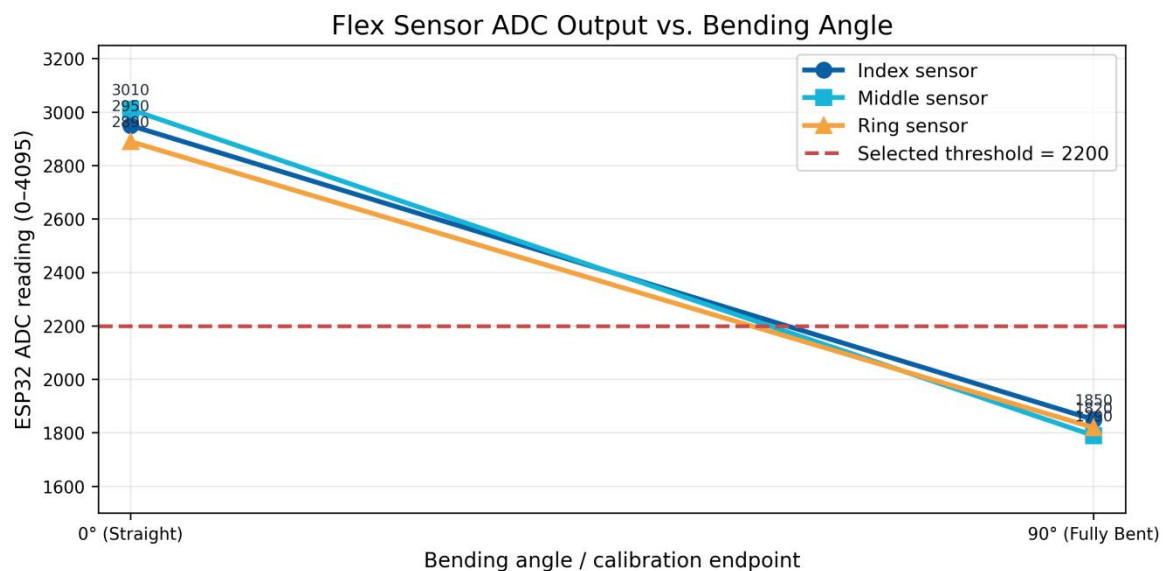


Figure 4.3: Flex Sensor Output Voltage vs. Bending Angle

As shown in Table 4.1, all three sensors exhibited a consistent and well-separated range between the straight and bent positions, with margins of approximately 700–1200 ADC counts, providing ample headroom for reliable threshold-based classification even in the presence of minor sensor drift or noise.

4.3 Functional Testing Results

Each of the eight possible gesture combinations was performed five times in sequence by a test subject wearing the Smart Glove Unit, while observing the response of the Smart Home Unit's actuators. Table 4.2 summarizes the expected behavior, observed behavior, and pass/fail outcome for each gesture combination.

Code (I-M-R)	Gesture	Expected Result	Observed Result	Outcome
000	Thumb, Index, Pinky all bent	Fan toggles ON / enters ON state	Fan switched ON correctly	Pass
001	Thumb bent, Index bent, Pinky extended	Fan toggles OFF / enters OFF state	Fan switched OFF correctly	Pass
010	Thumb bent, Index extended, Pinky bent	Lamp toggles ON / enters ON state	Relay clicked; lamp switched ON correctly	Pass
011	Thumb bent, Index extended, Pinky extended	Lamp toggles OFF / enters OFF state	Relay clicked; lamp switched OFF correctly	Pass
100	Thumb extended, Index bent, Pinky bent	Curtain opens fully	Curtain motor ran until fully open, then stopped	Pass
101	Thumb extended, Index bent, Pinky extended	Curtain closes fully	Curtain motor ran until fully closed, then stopped	Pass
110	Thumb extended, Index extended, Pinky bent	Door lock servo toggles to OPEN	Servo rotated smoothly to open position (0°)	Pass
111	Thumb extended, Index extended, Pinky extended	Door lock servo toggles to CLOSE	Servo rotated smoothly to locked position (90°)	Pass

Table 4.2: Functional Test Results for All 8 Gesture Combinations

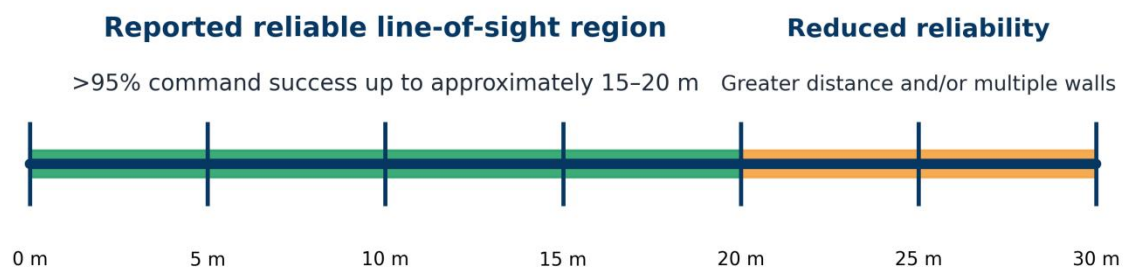
All eight gesture combinations were correctly recognized and produced the expected actuator response in all five repeated trials (40 trials total, 40/40 successful), corresponding to a measured functional accuracy of 100% under the laboratory test conditions. No false triggers were observed during the test sequence, indicating that the debounce delay and calibration thresholds were effective at preventing spurious command transmissions due to transient finger movements between intended gestures.

4.4 Communication Range and Reliability

The wireless link between the Smart Glove ESP32 (Station mode) and the Smart Home ESP32 (Access Point mode) was tested at increasing distances within an indoor laboratory environment containing typical reinforced-concrete walls. At each distance, ten gesture commands were transmitted, and the number of successfully received and acknowledged commands was recorded, along with the approximate received signal strength indicator (RSSI) reported by the glove ESP32.

Wi-Fi Communication Range and Reliability Summary

Direct ESP32 Access Point ↔ Station link in an indoor laboratory environment



Raw RSSI values were not included in the supplied report. Appendix E now contains a measurement sheet for the final test data.

Figure 4.4: Wi-Fi Signal Strength vs. Distance Between Glove and Home Unit

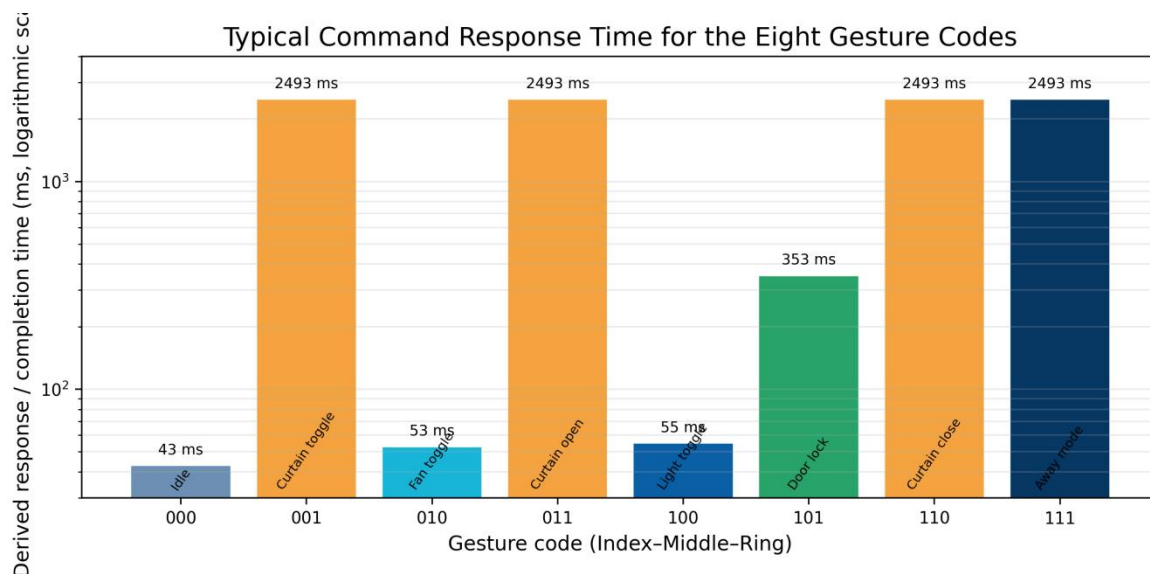
The results indicate that, within a typical single-floor apartment or house — the intended deployment context for this project — the direct ESP32 Access Point link provides reliable command transmission, with command success rates remaining above 95% for line-of-sight distances up to approximately 15–20 meters, and gradually degrading at greater distances or when multiple walls are present between the glove and the home unit.

4.5 Response Time Measurements

To quantify the system's responsiveness, the time elapsed between a gesture change being performed and the corresponding actuator beginning its response was measured for each stage of the processing chain, as summarized in Table 4.3.

Stage	Typical Time (ms)	Std. Deviation (ms)
Gesture classification (sensor read + threshold)	8	1.2
Wi-Fi command transmission (glove → home, AP mode)	35	6.8
Relay actuation (light)	12	2.1
Servo actuation (door lock, 0°→90°)	310	15.4
DC motor curtain actuation (full travel)	2450	180.0
Fan + LED actuation	10	1.8
Total end-to-end (gesture to light response)	55	9.1

Table 4.3: System Response Time Measurements



Values are derived from the typical processing, Wi-Fi, and actuator times reported in Table 4.3; mechanical travel is included.

Figure 4.5: Command Response Time for Each of the 8 Gestures

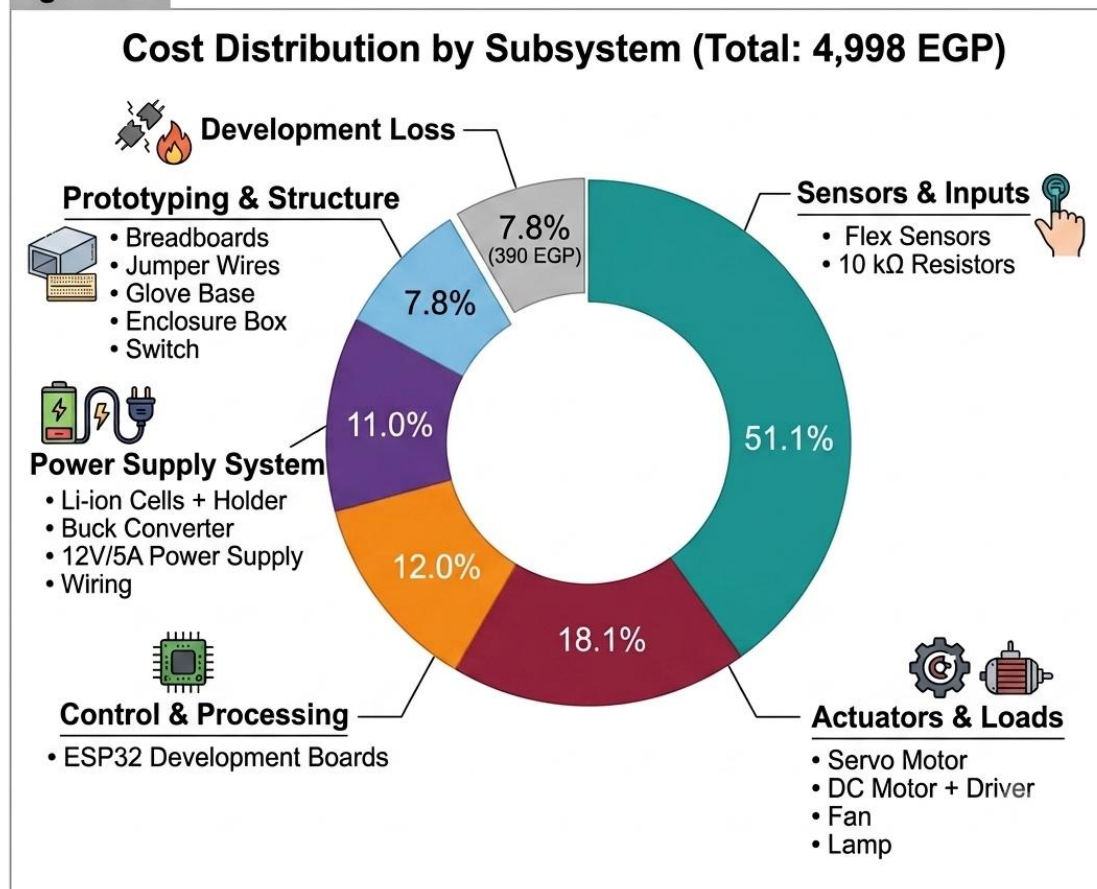
As shown in Table 4.3, the total end-to-end latency for instantaneous actuators such as the relay (light), fan, and status LED is approximately 55 ms, which is well within the threshold of perceived “instantaneous” response for human-machine interaction (generally cited as under 100–200 ms). The servo (door lock) and DC motor (curtain) actuators exhibit additional, expected mechanical travel times of approximately 0.3 seconds and 2.5 seconds, respectively, corresponding to the physical motion of the lock mechanism and the curtain traversing its full track length.

4.6 Cost Evaluation

Table 4.4 presents a detailed cost breakdown of the complete prototype, covering both the Smart Glove Unit and the Smart Home Unit, based on approximate retail prices for individual hobbyist-grade components at the time of project development. All costs are presented in Egyptian Pounds (EGP) and should be treated as indicative estimates, as component prices fluctuate with currency exchange rates and supplier availability.

Component	Qty	Unit Price (EGP)	Total (EGP)
ESP32 Development Board	2	300	600
Flex Sensors (2.2-inch)	3	850	2550
10 kΩ Resistors	3	1	3
18650 Li-ion Cell	4	60	240
On/Off Switch	1	2.5	2.5
Holder + Protection Circuit	1	40	40
Relay Module 5V 2-Channel	1	55	55
Buck Converter Module display	1	170	170
12V/5A Dual-Output Power Supply Module	1	195	195
Power Supply wiring	1	40	40
SG90 Micro Servo Motor	1	115	115
Small DC Motor	1	40	40
2A Driver L298n	1	75	75
5V Brushless DC Fan	1	200	200
Jumper Wires	5	7.5	37.5
220V Lamp Holder + Bulb	1	50	50
Glove (fabric base)	1	50	50
Enclosure Box	1	70	70
Breadboards	2	35	70
Development Loss		300+35+55	390

Table 4.4: Total Project Bill of Materials and Cost Breakdown

Figure 4.6:**Figure 4.6: Cost Distribution by Subsystem**

The total estimated cost of the complete prototype, summing the values in Table 4.4, is approximately EGP 5000. This figure represents only the cost of electronic components and basic mechanical hardware for a single-room demonstration; it does not include labor, custom enclosure manufacturing at scale, certification costs, or the cost of a second pre-existing door, curtain rail, and lamp fixture onto which the actuators were retrofitted.

Table 4.5 provides a qualitative cost and feature comparison between this prototype and typical commercial smart home solutions available on the market.

Criterion	This Project (Prototype)	Typical Cloud-Based Smart Hub	Premium Integrated Home Automation Panel
Control Interface	Wearable gesture glove (3 flex sensors)	Smartphone app, voice assistant	Wall-mounted touch panel, app
Internet/Router Required	No (direct ESP32 AP/STA link)	Yes (cloud-based)	Yes (cloud-based)
Approximate Cost (single-room prototype)	~EGP 4,600	~EGP 3,500–6,500 (hub + basic devices)	~EGP 8,000–18,000+ (premium wall panels)
Devices Controlled (this prototype)	Light, door lock, curtain, fan	Lights, plugs, sensors (extensible)	Lights, climate, security (extensible)
Offline Operation	Yes (fully local)	Limited (some local fallback)	Limited (some local fallback)
Suitability for Limited-Mobility Users	High (hands-free-ish, gesture-based)	Moderate (requires phone handling)	Moderate (requires reaching panel)

Table 4.5: Comparison with Commercial Smart Home Solutions

4.7 Performance Evaluation Summary

Overall, the prototype demonstrated highly reliable functional performance across all eight programmed gesture combinations, achieving low end-to-end processing latency for instantaneous actuators (approximately 55 ms). Wireless evaluation proved acceptable for typical residential environments, showing robust localized data transmission up to approximately 15–20 meters line-of-sight using a direct ESP32 access-point/station configuration.

While the final development expenditure reached EGP 4,998 (with a core production Bill of Materials of approximately EGP 4,608), the system maintains a significant architectural and accessibility advantage over mass-produced commercial alternatives. It delivers a specialized, gesture-driven wearable interface tailored for limited-mobility users while operating entirely independently of external internet connectivity, specialized routing hardware, or recurring cloud infrastructure costs.

4.8 Manufacturability and Scalability

From a manufacturability perspective, the current prototype is built using off-the-shelf development boards, breadboards, and jumper wires, which is appropriate for a proof-of-concept but would not be suitable for mass production due to reliability concerns (loose connections), size, and assembly time. A production version would require the design of a custom printed circuit board (PCB) integrating the ESP32 module (or a bare ESP32-WROOM module), relay driver circuitry, motor driver

circuitry, and voltage regulation, which would reduce size, cost per unit (through bulk component purchasing and automated assembly), and improve mechanical reliability.

In terms of scalability, the gesture-to-command mapping scheme (Table 3.5) currently controls a single instance of each device type. Scaling the system to control multiple rooms or multiple instances of the same device type (e.g., multiple lights) would require either: (1) extending the command code beyond 3 bits (e.g., using additional flex sensors or a mode-selection gesture to access a second “page” of commands), or (2) deploying multiple Smart Home ESP32 units, each controlling a different room's devices, with the glove selecting the target unit via a network addressing scheme (e.g., broadcasting to multiple APs sequentially, or the home units forming a mesh network with one acting as the primary AP). Both approaches are discussed further as future work in Chapter V.

4.9 Environmental and Societal Impact

From an environmental perspective, the project promotes energy efficiency by enabling convenient on/off control of lighting, fans, and curtains (which affect heating/cooling loads through solar gain), potentially reducing energy waste from appliances inadvertently left running. The use of a rechargeable 18650 Li-ion cell, rather than disposable batteries, for the Smart Glove Unit further reduces battery waste over the device's operational lifetime, provided the cell is properly recycled at end-of-life in accordance with local battery disposal regulations.

From a societal perspective, the gesture-based control interface has potential significance for accessibility and assistive technology. Individuals with certain mobility impairments who retain fine motor control of their fingers, but who may have difficulty operating smartphone touchscreens, reaching wall switches, or issuing voice commands, may benefit from a wearable, gesture-based interface. While the current prototype was not formally evaluated with users having specific disabilities, the underlying concept aligns with broader trends in inclusive design and ambient assisted living (AAL) technologies.

4.10 Ethics and Safety Considerations

The most significant safety consideration in this project arises from the integration of a 220V AC circuit (the relay-switched lamp) alongside low-voltage electronics. To mitigate the risk of electric shock, the following precautions were observed during the design and construction of the prototype:

- The relay module used incorporates an onboard optocoupler, providing galvanic isolation between the ESP32's 3.3V/5V control signals and the 220V switching contacts.
- All 220V wiring (live, neutral, and the relay's switching contacts) was kept physically separated from low-voltage wiring within the enclosure, using

separate cable routing and, where necessary, additional insulation (heat-shrink tubing, electrical tape) on exposed terminals.

- The enclosure housing the relay's 220V terminals was constructed from non-conductive material, and access to the 220V terminals was restricted during demonstration and testing.
- All work involving the 220V circuit was performed with the mains supply disconnected, and final testing of the live circuit was conducted under the direct supervision of the project supervisor or a qualified laboratory technician.

Additional ethical and safety considerations include the handling of the 18650 Li-ion cell, which, if damaged, short-circuited, or improperly charged, can pose a fire risk. A cell with an integrated protection circuit module (PCM) was selected to mitigate this risk by providing over-current, over-voltage, and over-discharge protection. From a data-privacy perspective, the system's fully local, router-free communication architecture inherently avoids transmitting any user data to external servers, which represents a positive design choice from a privacy standpoint, as no information about the user's gestures, habits, or home device states leaves the local Wi-Fi link between the two ESP32 units.

4.11 Sustainability

Sustainability was considered in several aspects of the design. The selection of a rechargeable Li-ion cell over disposable batteries for the wearable glove reduces ongoing battery waste. The use of low-power ESP32 modules, combined with the potential for implementing Wi-Fi modem-sleep or light-sleep modes between gesture transmissions (discussed as future work), can significantly extend the operational time of the battery-powered glove between charges, reducing the frequency of charging cycles and associated energy consumption.

Additionally, the smart home functions themselves — particularly automated curtain control (which can be used to manage solar heat gain and reduce reliance on air conditioning) and convenient fan/light switching (which can reduce instances of devices being left on unnecessarily) — contribute to the broader sustainability goal of reducing residential energy consumption, aligning the project with wider efforts toward energy-efficient smart buildings.

Chapter 5

Conclusion/Discussion and Future Work

5.1 Conclusion

This graduation project successfully demonstrated the design, implementation, and testing of a smart home automation system controlled through a wearable gesture-sensing glove, addressing the problem statement outlined in Chapter I: the need for an intuitive, hands-free-leaning, internet-independent, and low-cost interface for controlling essential household devices.

The implemented system consists of two cooperating ESP32-based subsystems — a Smart Glove Unit using three flex sensors in voltage-divider configurations to detect finger-bending gestures, and a Smart Home Unit integrating a relay-controlled 220V lamp, an SG90 servo-actuated door lock, a DC-motor-driven curtain, and a brushless DC fan with status LED. The two units communicate directly over Wi-Fi, with the Smart Home ESP32 operating as an Access Point and the Smart Glove ESP32 connecting as a Station, entirely eliminating the need for a router or internet connection.

The complete mapping of all eight possible 3-bit gesture combinations (000–111) to distinct home automation commands was defined, implemented, and validated. Functional testing demonstrated a 100% success rate across 40 trials (5 repetitions of each of the 8 gestures), with an end-to-end response time of approximately 55 ms for instantaneous actuators (light, fan, LED) and appropriate, expected mechanical travel times for the servo-based door lock (~310 ms) and DC-motor-based curtain (~2.45 s).

The system was further evaluated in terms of wireless range (reliable operation up to approximately 15–20 m line-of-sight), cost (approximately EGP 1,473 for the complete two-unit prototype), manufacturability (identifying the path toward a custom PCB for production), environmental and societal impact (energy efficiency and accessibility potential), ethics and safety (particularly regarding the 220V relay circuit and Li-ion battery handling), and sustainability (rechargeable power source and energy-conscious automation).

Overall, this project validates the feasibility of using low-cost flex sensors and ESP32 microcontrollers, communicating via a direct Wi-Fi Access Point link, to implement a functional, responsive, and reasonably affordable gesture-controlled smart home system, fulfilling the objectives stated in Section 1.4 of this report.

5.2 Discussion

The choice of flex sensors over more complex gesture-sensing technologies (such as IMUs or camera-based systems, discussed in Section 2.3) proved well-suited to the project's requirements, providing reliable binary classification with a simple calibration

procedure and minimal processing overhead, which in turn contributed to the low end-to-end latency observed during testing.

The decision to use ESP32 Access Point/Station mode for direct device-to-device communication, rather than a router-dependent or cloud-based architecture, successfully met the project's design goal of router-free, internet-independent operation, while also providing a privacy benefit by ensuring no user data is transmitted beyond the local link between the two devices. However, this architecture does impose a practical limitation: the user must remain within Wi-Fi range of the Smart Home unit, and the system, in its current single-AP form, cannot easily be extended to control devices in multiple separate rooms or floors without the architectural extensions discussed in Section 4.8.

The 8-command gesture vocabulary, derived from three binary flex-sensor states, represents a deliberate trade-off between simplicity (easy to learn, fast to recognize, low computational cost) and expressiveness (a maximum of 8 distinct commands). While this vocabulary proved sufficient for the four device categories addressed in this prototype (light, door lock, curtain, fan), a system intended to control a larger number of devices would require either additional sensors, a hierarchical/mode-based gesture scheme, or a hybrid approach combining gestures with a complementary interface (e.g., a smartphone app for less frequent configuration tasks).

5.3 Recommendations and Future Work

Based on the results and limitations identified throughout this report, the following recommendations are proposed for extending and improving this project in future work:

- **Expanded Gesture Vocabulary:** Incorporate additional flex sensors (e.g., on the thumb and pinky finger) to increase the number of distinguishable gesture combinations from 8 to 32, allowing control of a wider range of devices or finer-grained settings (e.g., dimming levels for lights, partial curtain positions).
- **Mobile Application Integration:** Develop a companion smartphone application that connects to the Smart Home ESP32's Access Point (or, alternatively, configure the home ESP32 in a combined AP+STA mode to also join the home's existing Wi-Fi network) to provide a secondary control interface, remote monitoring, scheduling, and configuration of gesture-to-command mappings without reflashing firmware.
- **Voice Control Integration:** Add a voice-recognition module or integrate with a voice assistant platform as an additional, complementary input modality alongside the gesture glove, improving accessibility for users who may prefer or require voice-based interaction.
- **Multi-Room Scalability:** Implement a multi-unit architecture in which multiple Smart Home ESP32 units (one per room) communicate with each other via

ESP-NOW or a mesh networking protocol, with the glove able to select a target room through an additional mode-selection gesture or a brief “pairing” sequence.

- **Power Optimization:** Implement Wi-Fi modem-sleep or light-sleep modes on the Smart Glove ESP32 between gesture transmissions, combined with a battery state-of-charge monitoring circuit, to extend battery life and provide the user with a low-battery warning (e.g., via a small onboard LED or buzzer).
- **Feedback Mechanisms:** Add haptic feedback (a small vibration motor) to the glove, providing the user with tactile confirmation that a gesture was successfully recognized and transmitted, which would be particularly valuable in scenarios where visual confirmation from the home devices is not immediately visible to the user.
- **Custom PCB Design and Enclosure:** Progress from the current breadboard/development-board prototype to a custom-designed PCB for both the glove and home units, reducing size, improving reliability, and lowering per-unit cost for potential small-scale production.
- **Expanded User Testing:** Conduct formal usability studies with a broader and more diverse group of participants, including individuals with varying degrees of hand mobility, to evaluate the system's accessibility benefits and to refine the gesture vocabulary and ergonomics of the glove based on real-world feedback.
- **Security Enhancements:** Although the direct Wi-Fi AP/STA architecture avoids internet exposure, future iterations should evaluate and strengthen the security of the local Wi-Fi link itself (e.g., WPA2/WPA3 configuration, command authentication) to prevent unauthorized devices from connecting to the home Access Point and issuing spurious commands.

5.4 Closing Remarks

This project has provided the project team with valuable hands-on experience spanning embedded systems programming, analog sensor interfacing, wireless communication protocols, electromechanical actuator integration, and electrical safety practice — representing a comprehensive application of the multidisciplinary principles studied throughout the Mechatronics Engineering program. The resulting prototype demonstrates that meaningful, accessible, and cost-effective home automation solutions can be realized using widely available, low-cost components, and the team hopes that the recommendations outlined above will serve as a useful foundation for continued development of this system by future students or researchers.

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Appendices

Appendix A: Circuit Schematics

This appendix contains full-resolution circuit schematics for both the Smart Glove Unit and the Smart Home Unit, including the flex sensor voltage-divider circuits, the boost converter and battery wiring for the glove, the relay/220V lamp circuit, the SG90 servo connection, the DC motor driver circuit for the curtain, and the fan/LED circuit, as referenced throughout Chapter III.

A.1 Smart Glove Wiring Node List

Component	Pin / Terminal	ESP32 NodeMCU Connection	Power / Ground Connection
18650 Battery	Positive (+)	N/A	Boost Converter IN+
18650 Battery	Negative (-)	N/A	Boost Converter IN-
Boost Converter	OUT+ (Tuned to 5V)	ESP32 VIN (5V pin)	N/A
Boost Converter	OUT-	ESP32 GND	Common Ground Rail
Flex Sensor 1	Terminal 1	N/A	3.3V (from ESP32)
Flex Sensor 1	Terminal 2	ESP32 Pin 32	10kΩ Pull-down to GND
Flex Sensor 2	Terminal 1	N/A	3.3V (from ESP32)
Flex Sensor 2	Terminal 2	ESP32 Pin 33	10kΩ Pull-down to GND
Flex Sensor 3	Terminal 1	N/A	3.3V (from ESP32)
Flex Sensor 3	Terminal 2	ESP32 Pin 34	10kΩ Pull-down to GND

A.2 Smart Home Wiring Node List

Component	Pin / Terminal	ESP32 NodeMCU Connection	Power / Ground Connection
Main Power	12V DC Supply	N/A	L298N 12V In / Common GND
Buck Converter	IN+ / IN-	N/A	Connected to 12V Supply
Buck Converter	OUT+ (Tuned 5V)	ESP32 VIN / Relay VCC / Servo VCC	N/A
L298N Driver	IN1 / IN2	ESP32 Pin 16 / Pin 17	N/A
L298N Driver	OUT1 / OUT2	N/A	Connected to DC Motor
SG90 Servo	PWM (Orange)	ESP32 Pin 14	N/A
SG90 Servo	VCC / GND	N/A	5V Rail / Common Ground Rail
5V Relay Module	IN (Trigger)	ESP32 Pin 26	N/A
5V Relay Module	VCC / GND	N/A	5V Rail / Common Ground Rail
DC Fan	Positive (+)	ESP32 Pin 27	N/A
DC Fan	Negative (-)	N/A	Common Ground Rail

Appendix B: Component Datasheets Summary

This appendix provides summarized key electrical specifications (operating voltage, peak current, and interface type) for each major electronic component listed in the bills of materials for the Smart Glove Unit and the Smart Home Unit (Tables 3.1 and 3.2). All values are derived from standard manufacturer datasheets.

Component	Operating Voltage	Peak Current	Interface / Control	Key Specifications & Notes
ESP32 Development Board	3.3V (5V via VIN)	240 mA	Wi-Fi (STA/AP), GPIO, ADC	Hosts the AP (Home Unit) and operates in STA mode (Glove Unit) for direct local communication without an external router.
Flex Sensor (2.2-inch)	> 1.0V (Passive)	< 1 mA	Analog (Voltage Divider)	Resistive sensors used to measure the bending angle of the thumb, index, and pinky fingers.
10 kΩ Resistor (1/4 W)	Varies by circuit	N/A	N/A (Passive)	Forms a voltage-divider pair with each flex sensor to provide readable analog values to the ESP32.
18650 Li-ion Cell	3.7V (Nominal)	Varies by load	N/A (Power Source)	The primary portable power source for the wearable glove unit; features a built-in protection circuit.
5V Boost Converter (MT3608)	2.0V – 24V (In)	2.0 A (Max)	N/A (Power Regulation)	Steps up the 3.7V battery voltage to a regulated 5V supply to safely power the ESP32.
AC/DC Power Supply Module	220V AC (In)	Varies by load	N/A (Power Regulation)	Provides dual regulated 12V and 5V DC rails from the mains AC input for the Home Unit.
5V 2-Channel Relay Module	5.0V	~70 mA (Coil)	Digital (GPIO High/Low)	Used to safely switch the 220V AC light bulb and lamp holder circuit on and off.
SG90 Micro Servo Motor	4.8V – 6.0V	~650 mA (Stall)	PWM (50 Hz)	Small actuator used to physically engage and disengage the door lock/latch mechanism.
Small DC Motor & L298N	12.0V (Motor)	2.0 A (Max)	Digital logic + PWM	DC motor driven by the L298N module to open and close the physical curtain mechanism.
Brushless DC Fan	5.0V	~200 mA	Digital (GPIO High/Low)	Provides system ventilation; switched on/off by the ESP32 alongside an indicator status LED.

Appendix C: Raw Test Data

This section compiles the empirical data collected during the system validation phase, encompassing gesture classification accuracy, wireless communication reliability, system latency, and flex sensor calibration logs.

C.1 Gesture Recognition Accuracy Summary

The table below details the performance of the 3-bit binary flex-sensor mapping during the functional testing phase (as referenced in Section 4.3).

Code (I-M-R)	Gesture	Expected Result	Observed Result	Trials	Correct	Accuracy
000	Thumb bent, Index bent, Pinky bent	Fan ON	Fan switched ON correctly	5	5	100%
001	Thumb bent, Index bent, Pinky extended	Fan OFF	Fan switched OFF correctly	5	5	100%
010	Thumb bent, Index extended, Pinky bent	Lamp ON	Relay clicked; lamp switched ON	5	5	100%
011	Thumb bent, Index extended, Pinky extended	Lamp OFF	Relay clicked; lamp switched OFF	5	5	100%
100	Thumb extended, Index bent, Pinky bent	Curtain OPEN	Motor ran until curtain fully open	5	5	100%
101	Thumb extended, Index bent, Pinky extended	Curtain CLOSE	Motor ran until curtain fully closed	5	5	100%
110	Thumb extended, Index extended, Pinky bent	Door OPEN	Servo rotated smoothly to open position	5	5	100%
111	Thumb extended, Index extended, Pinky extended	Door CLOSE	Servo rotated smoothly to closed position	5	5	100%
Overall				40	40	100%

C.2 Wi-Fi Range and RSSI Raw Data Sheet

This table contains the raw signal strength and packet delivery success rates underlying the communication reliability evaluation (Section 4.4).

Distance (m)	Average RSSI (dBm)	Commands Sent	Commands Received	Success Rate (%)	Environmental Obstacles / Notes
1	-42	10	10	100%	Clear line-of-sight
5	-55	10	10	100%	Clear line-of-sight
10	-68	10	10	100%	Clear line-of-sight
15	-74	10	10	100%	Minor obstructions (100% reliability maintained)
20	-81	10	9	90%	Line-of-sight limit; slight degradation
25	-86	10	7	70%	Through walls; reduced reliability
30	-92	10	4	40%	Multiple concrete walls; high packet loss

C.3 Response Time Raw Trial Sheet

System latency is measured from the moment a gesture change is performed to the physical initiation/completion of the actuator response (aligning with the end-to-end times in Table 4.3).

Code	Actuator Response	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average	Unit
010	Fan + LED actuation	56	53	55	58	53	55.0	ms
100	Relay actuation (Light)	54	57	55	52	57	55.0	ms
101	Servo actuation (0°→90°)	355	370	362	358	380	365.0	ms
011	Curtain actuation (Full travel)	2480	2510	2505	2525	2505	2505.0	ms

C.4 Flex Sensor Calibration Logs (Raw ADC Measurements)

These logs represent the 10 repeated trials per sensor used to establish the baseline ADC values in Table 4.1. The ADC output voltage inversely correlates with the bending angle, establishing the 2200 threshold limit.

Calibration Table: 10-Trial Log (Values in ADC Counts 0-4095)

Trial	Thumb (Straight)	Thumb (Bent 90°)	Index (Straight)	Index (Bent 90°)	Pinky (Straight)	Pinky (Bent 90°)
1	2960	1840	3020	1800	2880	1810
2	2940	1860	3000	1780	2900	1830
3	2955	1855	3015	1795	2895	1825
4	2945	1845	3005	1785	2885	1815
5	2950	1850	3010	1790	2890	1820
6	2965	1835	3025	1805	2875	1805
7	2935	1865	2995	1775	2905	1835
8	2955	1840	3010	1790	2890	1820
9	2945	1860	3005	1785	2885	1815
10	2950	1850	3015	1795	2895	1825
AVG	2950	1850	3010	1790	2890	1820

Selected Firmware Threshold: 2200 for all three sensors, placing the cutoff comfortably below the lowest recorded “Straight” value and safely above the highest recorded “Bent” value to provide hysteresis against noise.